

NANYANG TECHNOLOGICAL UNIVERSITY  
SEMESTER I EXAMINATION 2019-2020  
MH2500 – Probability and Introduction to Statistics

December 2019

TIME ALLOWED: 2 HOURS

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INSTRUCTIONS TO CANDIDATES

1. This examination paper contains **FIVE (5)** questions and comprises **FIVE (5)** printed pages including appendix.
2. Answer **ALL** questions. The marks for each question are indicated at the beginning of each question.
3. Answer each question beginning on a **FRESH** page of the answer book.
4. This is a **RESTRICTED OPEN BOOK** exam. Candidates are allowed **ONE** double-sided A4 size help sheet.
5. Candidates may use calculators. However, they should write down systematically the steps in the workings.
6. The table with the values of the standard normal distribution is included at the end of this examination paper.
7. All questions are the property of Nanyang Technological University.

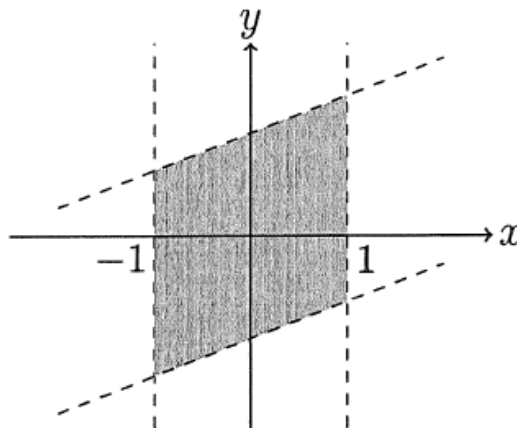
**QUESTION 1****(15 marks)**

An urn contains 4 blue balls and 3 red balls. A fair die is rolled. If the number shown on the die is  $i$  ( $i = 1, 2, \dots, 6$ ), then  $i$  balls are drawn from the urn uniformly at random without replacement.

- (a) (5 marks) When  $i = 3$ , what is the probability that all balls drawn are red?
- (b) (10 marks) Given that all balls drawn are red, what is the probability that the remaining balls in the urn are all blue?

## QUESTION 2

(30 marks)



For each  $\theta \in [0, \pi/2)$ , we define a planar region  $A_\theta$  as

$$A_\theta = \{f(x, y) : |x \cos \theta - y \sin \theta| < 1\}, \quad 0 \leq \theta < \frac{\pi}{2}.$$

For each  $\theta \in (0, \pi/2)$ , we define a planar region  $D_\theta$  as

$$D_\theta = A_0 \cap A_\theta, \quad 0 < \theta < \frac{\pi}{2}.$$

The shaded region on the right is a sketch of  $D_{6/5}$ .

Let  $(X, Y)$  be a point uniformly distributed in  $D_\theta$  for some  $\theta \in (0, \pi/2)$ .

- (5 marks) Find the marginal distribution of  $X$ .
- (5 marks) Find the conditional density function of  $Y$  given  $X = x$ .
- (5 marks) Find the conditional expectation  $\mathbb{E}(Y|X)$ .
- (5 marks) Find  $\mathbb{E}(Y)$ .
- (5 marks) Find  $\text{Var}(Y)$ .
- (5 marks) Are  $X$  and  $Y$  correlated? If they are correlated, are they positively or negatively correlated? If they are not correlated, are they independent? State your reason.

**QUESTION 3****(20 marks)**

Suppose that a fair die is rolled  $n$  times independently. For each  $i = 1, \dots, 6$ , let  $X_i$  denote the number of times  $i$  appears.

- (a) (2 marks) Find the probability mass function of  $X_2$ .
- (b) (2 marks) Find the joint probability mass function of  $(X_2, X_4)$ .
- (c) (2 marks) Find the conditional probability mass function of  $X_2$  given that  $X_4 = l$  ( $0 \leq l \leq n$ ).
- (d) (4 marks) Find the probability mass function of  $X_2 + X_4$ .
- (e) (10 marks) Find  $\text{Cov}(X_2, X_4)$ .

**QUESTION 4****(20 marks)**

For each integer  $m \geq 1$ , we define  $2m$  numbers  $b_{m,1}, \dots, b_{m,2m}$  as follows:

$$b_{m,i} = \begin{cases} 1, & 1 \leq i \leq m; \\ -1, & m+1 \leq i \leq 2m. \end{cases}$$

Let  $Y_1, Y_2, \dots$  be i.i.d. random variables such that  $\Pr\{Y_i = 0\} = \Pr\{Y_i = 1\} = 1/2$ .

- (a) (10 marks) Show that there exists a constant  $c_1 > 0$  (which does not depend on  $m$ ) such that for all  $m \geq 1$ ,

$$\Pr \left\{ \left| \sum_{i=1}^{2m} b_{m,i} Y_i \right| > c_1 \sqrt{m} \right\} \leq \frac{1}{10}.$$

- (b) (10 marks) Show that there exists a constant  $c_2 > 0$  (which does not depend on  $m$ ) such that for all  $m \geq 1$ ,

$$\lim_{n \rightarrow \infty} \Pr \left\{ \left| \sum_{i=1}^{2m} b_{m,i} Y_i \right| > c_2 \sqrt{m} \right\} > \frac{9}{10}.$$

Note: Question 5 (Hypothesis Testing) is not tested for the Final Exam.

**END OF PAPER**